



DAFFODIL INTERNATIONAL UNIVERSITY
Department of Computing and Information System (CIS)

Algorithm

Lab Report-05

Submitted By:

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Program: B.Sc. in CIS

Submitted To:

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Class_Code

```
#include <stdio.h>

void merge(int arr[], int l, int m, int h)
{
    int i = l, j = m + 1, k = 0;
    int temp[h-l+1];

    while(i <= m && j <= h)
    {
        if(arr[i] < arr[j])
            temp[k++] = arr[i++];
        else
            temp[k++] = arr[j++];
    }
    while(i <= m)
        temp[k++] = arr[i++];

    while(j <= h)
        temp[k++] = arr[j++];

    for(i = l, k = 0; i <= h; i++, k++)
        arr[i] = temp[k];
}

void mergeSort(int arr[], int l, int h)
{
    if(l < h)
    {
        int m = (l + h) / 2;

        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, h);

        merge(arr, l, m, h);
    }
}

int main()
{
    int n, arr[50];

    printf("Enter Size: ");
    scanf("%d", &n);

    printf("Enter Elements: ");
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    mergeSort(arr, 0, n-1);
```

```
printf("Sorted Array: ");
for(int i = 0; i < n; i++)
    printf("%d ", arr[i]);

return 0;
}
```

Output:

```
{ gcc classCode.c -o classCode } ; if ($?) { .\classCode }
Enter Size: 5
Enter Elements: 55
44
66
77
11
Sorted Array: 11 44 55 66 77
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```

Task 1

```
#include <stdio.h>

void merge(int arr[], int l, int m, int h)
{
    int i = l, j = m + 1, k = 0;
    int temp[h-l+1];

    while(i <= m && j <= h)
    {
        if(arr[i] < arr[j])
            temp[k++] = arr[i++];
        else
            temp[k++] = arr[j++];
    }
    while(i <= m)
        temp[k++] = arr[i++];

    while(j <= h)
        temp[k++] = arr[j++];

    for(i = l, k = 0; i <= h; i++, k++)
        arr[i] = temp[k];
}

void mergeSort(int arr[], int l, int h)
{
    if(l < h)
```

```
{
    int m = (l + h) / 2;

    mergeSort(arr, l, m);
    mergeSort(arr, m + 1, h);

    merge(arr, l, m, h);
}

int main()
{
    int n, marks[100];

    printf("Enter the number of students: ");
    scanf("%d", &n);

    printf("Enter the marks:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", &marks[i]);

    mergeSort(marks, 0, n-1);

    printf("\nSorted Marks (Ascending):\n");

    for(int i = 0; i < n; i++)
        printf("%d ", marks[i]);

    return 0;
}
```

Output:

```
{ gcc task_1.c -o task_1 } ; if ($?) { .\task_1 }
Enter the number of students: 4
Enter the marks:
88
78
91
85

Sorted Marks (Ascending):
78 85 88 91
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```

Task 2

```
#include <stdio.h>

void merge(int arr[], int l, int m, int h)
{
    int i = l, j = m + 1, k = 0;
    int temp[h-l+1];
    while(i <= m && j <= h)
    {
        if(arr[i] > arr[j])
            temp[k++] = arr[i++];
        else
            temp[k++] = arr[j++];
    }
    while(i <= m)
        temp[k++] = arr[i++];

    while(j <= h)
        temp[k++] = arr[j++];

    for(i = l, k = 0; i <= h; i++, k++)
        arr[i] = temp[k];
}

void mergeSort(int arr[], int l, int h)
{
    if(l < h)
    {
        int m = (l + h) / 2;

        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, h);

        merge(arr, l, m, h);
    }
}

int main()
{
    int n, order[100];
    printf("Enter the number of orders: ");
    scanf("%d", &n);
    printf("Enter the order values:\n");

    for(int i = 0; i < n; i++){
        scanf("%d", &order[i]);
    }
    mergeSort(order, 0, n-1);
    printf("\nSorted Order Values (Descending):\n");
    for(int i = 0; i < n; i++)
        printf("%d ", order[i]);
}
```

```
printf("\n\nTop 3 Highest Order Values:\n");
if(n >= 3)
{
    for(int i = 0; i < 3; i++)
        printf("%d ", order[i]);
}
else
{
    for(int i = 0; i < n; i++)
        printf("%d ", order[i]);
}
return 0;
}
```

Output:

```
{ gcc task_2.c -o task_2 } ; if ($?) { .\task_2 }
```

```
Enter the number of orders: 5
```

```
Enter the order values:
```

```
1200
```

```
1400
```

```
1100
```

```
1300
```

```
1500
```

```
Sorted Order Values (Descending):
```

```
1500 1400 1300 1200 1100
```

```
Top 3 Highest Order Values:
```

```
1500 1400 1300
```

```
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```

End